Theoretical Analysis

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1 Unified Coupled Objective: One Coherent Model

We model the system by two controllable quantities:

$$\tau > 0 \quad \text{learner synchronization interval}$$
$$\Lambda \quad \text{actor arrival rate / experience rate.}$$

We minimize wall-clock time to reach a fixed target quality (reward, KL, etc.) by

$$F(\tau, \Lambda) = t_{\text{step}}(\tau) \cdot T_{\text{steps}}(\tau, \Lambda),$$

where

- $t_{\text{step}}(\tau)$ is wall time per learner update (per micro-step),
- $T_{\text{steps}}(\tau, \Lambda)$ is the number of learner updates needed to reach the target.

The feasible domain is

$$\mathcal{D} = \{(\tau, \Lambda) : \tau > 0,\ 0 < \Lambda < \mu_\ell(\tau)\},$$

where $\mu_\ell(\tau)$ is learner throughput.

2 Hardware Side: $\tau \mapsto t_{\text{step}}(\tau)$, $\tau \mapsto \mu_\ell(\tau)$

We assume

$$t_{\text{step}}(\tau) = B_0 + B_1 \tau^{-\gamma}, \qquad B_0, B_1 > 0, \quad \gamma > 0,$$

and define learner throughput

$$\mu_\ell(\tau) = \frac{1}{t_{\text{step}}(\tau)} = \frac{1}{B_0 + B_1 \tau^{-\gamma}}.$$

Basic properties:

- As $\tau \to 0^+$, $t_{\text{step}}(\tau) \to +\infty$ and $\mu_\ell(\tau) \to 0$.
- As $\tau \to +\infty$, $t_{\text{step}}(\tau) \to B_0$ and $\mu_\ell(\tau) \to 1/B_0$.
- $\mu_\ell(\tau)$ is strictly increasing in τ.

3 Learning Side: $T_{\text{steps}}(\tau, \Lambda)$

We postulate the structured (exponent) model

$$T_{\text{steps}}(\tau, \Lambda) = K_0 + K_\tau \tau^q + K_\Lambda \Lambda^{-p} + K_\Delta \big(\mu_\ell(\tau) - \Lambda\big)^{-s},$$

with parameters

$$K_0, K_\tau, K_\Lambda, K_\Delta > 0, \qquad q > 1, \quad p > 0, \quad s > 0.$$

Interpretation:

- K_0: baseline update count.
- $K_\tau \tau^q$: drift penalty from infrequent synchronization.

- $K_\Lambda \Lambda^{-p}$: variance penalty when actor rate is too low.
- $K_\Delta(\mu_\ell(\tau) - \Lambda)^{-s}$: staleness/queue penalty as $\Lambda \to \mu_\ell(\tau)^-$.

The Coupled Objective in Closed Form
Combining the above:

$$F(\tau, \Lambda) = \Big(B_0 + B_1\tau^{-\gamma}\Big)\Big[K_0 + K_\tau \tau^q + K_\Lambda \Lambda^{-p} + K_\Delta\big(\mu_\ell(\tau) - \Lambda\big)^{-s}\Big],$$

subject to $(\tau, \Lambda) \in \mathcal{D}$, i.e. $\tau > 0$ and $0 < \Lambda < \mu_\ell(\tau)$, with $\mu_\ell(\tau) = 1/(B_0 + B_1\tau^{-\gamma})$.

4 Existence of an Interior Optimum

Proposition 1 (Existence of a minimizer)**.** *Assume $q > 1$, $\gamma > 0$, $p > 0$, $s > 0$, and all constants are positive. Then $F(\tau, \Lambda) \to +\infty$ on all boundary approaches of $\mathcal{D}$; hence F attains at least one global minimizer on $\mathcal{D}$, and any minimizer is interior.*

Proof sketch: Consider boundary cases:

1. $\tau \to 0^+$: $t_{\text{step}}(\tau) = B_0 + B_1\tau^{-\gamma} \to +\infty$, and $\mu_\ell(\tau) \to 0$. Since $0 < \Lambda < \mu_\ell(\tau)$, the feasible range collapses, and the staleness term $K_\Delta(\mu_\ell(\tau) - \Lambda)^{-s}$ explodes for any feasible Λ approaching the boundary; thus $F \to +\infty$.
2. $\tau \to +\infty$: $t_{\text{step}}(\tau) \to B_0$ but $K_\tau \tau^q \to +\infty$ because $q > 1$, thus $F \to +\infty$.
3. $\Lambda \to 0^+$: $K_\Lambda \Lambda^{-p} \to +\infty$, hence $F \to +\infty$.
4. $\Lambda \to \mu_\ell(\tau)^-$: $K_\Delta(\mu_\ell(\tau) - \Lambda)^{-s} \to +\infty$, hence $F \to +\infty$.

Therefore F is coercive toward $+\infty$ at the boundary of $\mathcal{D}$. Since F is continuous on $\mathcal{D}$, it attains a global minimum on some compact sublevel set contained in $\mathcal{D}$, yielding existence of an interior minimizer. □

5 Inner Problem: For Fixed τ, Unique Optimal $\Lambda^*(\tau)$

Fix τ, hence $\mu_\ell(\tau)$ is constant. Define the Λ-dependent part

$$\phi_\tau(\Lambda) := K_\Lambda \Lambda^{-p} + K_\Delta\big(\mu_\ell(\tau) - \Lambda\big)^{-s}, \qquad 0 < \Lambda < \mu_\ell(\tau).$$

Proposition 2 (Strict convexity and uniqueness of $\Lambda^{(\tau)}$)**.** *For fixed $\tau > 0$, ϕ_τ is strictly convex on $(0, \mu_\ell(\tau))$ and diverges at the endpoints. Hence it admits a unique minimizer $\Lambda^{(\tau)\in(0,\mu_\ell(\tau))}$.*

Proof. Compute derivatives:

$$\phi'_\tau(\Lambda) = -pK_\Lambda \Lambda^{-p-1} + sK_\Delta(\mu_\ell(\tau) - \Lambda)^{-s-1},$$

$$\phi''_\tau(\Lambda) = p(p+1)K_\Lambda \Lambda^{-p-2} + s(s+1)K_\Delta(\mu_\ell(\tau) - \Lambda)^{-s-2} > 0$$

on $(0, \mu_\ell(\tau))$, so ϕ_τ is strictly convex. As $\Lambda \to 0^+$, $\phi_\tau(\Lambda) \to +\infty$; as $\Lambda \to \mu_\ell(\tau)^-$, $\phi_\tau(\Lambda) \to +\infty$. Thus a unique minimizer exists. □

First-order optimality equation:
At the unique minimizer $\Lambda^*(\tau)$:

$$\phi'_\tau(\Lambda^*(\tau)) = 0 \quad \Longleftrightarrow \quad pK_\Lambda(\Lambda^*)^{-(p+1)} = sK_\Delta\big(\mu_\ell(\tau) - \Lambda^*\big)^{-(s+1)}.$$

Equivalently,

$$\big(\mu_\ell(\tau) - \Lambda^*\big)^{s+1} = K\,(\Lambda^*)^{p+1}, \qquad K = \frac{sK_\Delta}{pK_\Lambda} > 0.$$

6 Exact Hierarchical Decomposition

Define the reduced objective by plugging in the exact inner minimizer:

$$\Psi(\tau) := F(\tau, \Lambda^*(\tau)).$$

Proposition 3 (Exactness of the two-loop structure)**.** *Minimizing $F(\tau, \Lambda)$ over $(\tau, \Lambda) \in \mathcal{D}$ is equivalent to minimizing $\Psi(\tau)$ over $\tau > 0$. That is, any global minimizer (τ^*, Λ^*) satisfies $\Lambda^* = \Lambda^*(\tau^*)$ and $\tau^* \in \arg\min_{\tau>0} \Psi(\tau)$.*

Proof. For fixed τ, $t_{\text{step}}(\tau)$ is constant w.r.t. Λ, hence minimizing $F(\tau, \Lambda)$ over Λ is equivalent to minimizing $T_{\text{steps}}(\tau, \Lambda)$ over Λ, i.e. minimizing $\phi_\tau(\Lambda)$. Thus the unique minimizer is $\Lambda^*(\tau)$, and the joint minimization reduces to a 1D search over τ of $\Psi(\tau)$. □

7 General Power-Law Scaling of the Optimal Sync Interval

7.1 Abstract reduced model (uncoupled form used for scaling)

We consider the archetypal time-to-target objective (parameterized by a comm/comp ratio r):

$$T_{\text{total}}(\tau; r) = T_{\text{steps}}(\tau)\; t_{\text{step}}(\tau; r),$$

with the generalized exponent forms:

$$T_{\text{steps}}(\tau) \approx A_0 + A_1\tau^q, \qquad q > 1,$$

$$t_{\text{step}}(\tau; r) \approx B_0 + B_1 r\,\tau^{-\gamma}, \qquad \gamma > 0.$$

Thus

$$T_{\text{total}}(\tau; r) = (A_0 + A_1\tau^q)(B_0 + B_1 r\tau^{-\gamma}).$$

7.2 Stationary condition (explicit derivative)

Differentiate:

$$\frac{\partial}{\partial\tau} T_{\text{total}}(\tau; r) = A_1 q\tau^{q-1}(B_0 + B_1 r\tau^{-\gamma}) + (A_0 + A_1\tau^q)\big(B_1 r(-\gamma)\tau^{-\gamma-1}\big).$$

Set to zero:

$$A_1 q\tau^{q-1}(B_0 + B_1 r\tau^{-\gamma}) = \gamma B_1 r(A_0 + A_1\tau^q)\tau^{-\gamma-1}.$$

Multiply by $\tau^{\gamma+1}$:

$$A_1 q\tau^{q+\gamma}(B_0 + B_1 r\tau^{-\gamma}) = \gamma B_1 r(A_0 + A_1\tau^q).$$

Expand the left-hand side:

$$A_1 q\big(B_0\tau^{q+\gamma} + B_1 r\tau^q\big) = \gamma B_1 r A_0 + \gamma B_1 r A_1 \tau^q.$$

Rearrange the τ^q terms:

$$A_1 q B_0 \tau^{q+\gamma} + A_1 q B_1 r \tau^q - \gamma B_1 r A_1 \tau^q = \gamma B_1 r A_0,$$

$$A_1 q B_0 \tau^{q+\gamma} + A_1 B_1 r (q-\gamma)\tau^q = \gamma B_1 r A_0.$$

7.3 Asymptotic scaling

In the regime where the dominant balance is between the increasing drift term and decreasing amortized comm term (the regime motivating the scaling law), one obtains

$$\tau^*(r) \sim C\, r^{\frac{1}{q+\gamma}}.$$

Equivalently, a power-law:

$$\tau^*(r) \propto r^{\alpha}, \qquad \alpha = \frac{1}{q+\gamma}.$$

7.4 Special case: cube-root law

In convex delayed-SGD, we take $q = 2$ and the simple amortization model gives $\gamma = 1$, hence

$$\alpha = \frac{1}{2+1} = \frac{1}{3}, \qquad \Rightarrow \qquad \tau^* \propto r^{1/3}.$$

8 Communication Regime-Change Model (Overlap Knee)

8.1 AllReduce time model

Let one synchronization (one AllReduce) cost be

$$t_{\mathrm{AR}} = \underbrace{\alpha}_{\text{latency}} + \underbrace{\beta S}_{\text{bandwidth term}},$$

where S is the communicated size (bytes), α latency, β inverse bandwidth.

If syncing every τ steps, amortized communication per step (no overlap) is

$$t_{\mathrm{comm}}(\tau) = \frac{t_{\mathrm{AR}}}{\tau} = \frac{\alpha + \beta S}{\tau}.$$

8.2 Compute-comm overlap

Let t_{comp} be compute time per local step. Let $0 \le \rho \le 1$ denote an effective overlap fraction; then over τ steps, the overlappable time budget is

$$t_{\mathrm{ov}}(\tau) = \rho\, \tau\, t_{\mathrm{comp}}.$$

Thus exposed comm per sync is

$$t_{\mathrm{exp}}(\tau) = \max\{0,\, t_{\mathrm{AR}} - t_{\mathrm{ov}}(\tau)\} = \max\{0,\, (\alpha + \beta S) - \rho\tau t_{\mathrm{comp}}\}.$$

Amortized exposed comm per step:

$$t_{\text{comm,exp}}(\tau) = \frac{t_{\text{exp}}(\tau)}{\tau} = \max\left\{0,\ \frac{\alpha + \beta S}{\tau} - \rho t_{\text{comp}}\right\}.$$

Total per-step time:

$$t_{\text{step}}(\tau) = t_{\text{comp}} + t_{\text{comm,exp}}(\tau).$$

8.3 Two regimes and the knee

- **Regime A (comm not fully hidden):** if $(\alpha + \beta S) > \rho\tau t_{\text{comp}}$, then

  $$t_{\text{step}}(\tau) = (1-\rho)t_{\text{comp}} + \frac{\alpha + \beta S}{\tau},$$

  which recovers the amortized $1/\tau$ structure and hence the power-law scaling of τ.

- **Regime B (comm fully hidden):** if $(\alpha + \beta S) \le \rho\tau t_{\text{comp}}$, then

  $$t_{\text{exp}}(\tau) = 0 \quad \Rightarrow \quad t_{\text{step}}(\tau) = t_{\text{comp}},$$

  so increasing τ yields no further wall-time benefit, while drift still increases with τ. Therefore τ plateaus near the smallest value that achieves full overlap.

The transition occurs at the *overlap knee*

$$\tau_{\text{knee}} = \frac{\alpha + \beta S}{\rho\, t_{\text{comp}}}.$$

9 Exponent Invariance Under Coupling and When It Breaks

We now formalize whether optimizing Λ changes the scaling exponent of $\tau^{(r)}$.

9.1 Coupled model with explicit r

Use

$$t_{\text{step}}(\tau; r) = B_0 + B_1 r \tau^{-\gamma}, \qquad \gamma > 0, \quad \mu(\tau; r) = \frac{1}{t_{\text{step}}(\tau; r)}.$$

Steps-to-target:

$$T_{\text{steps}}(\tau, \Lambda; r) = K_0 + K_\tau \tau^q + K_\Lambda \Lambda^{-p} + K_\Delta(\mu(\tau; r) - \Lambda)^{-s},$$

so

$$F(\tau, \Lambda; r) = t_{\text{step}}(\tau; r)\, T_{\text{steps}}(\tau, \Lambda; r).$$

Define the inner minimizer

$$\Lambda^*(\tau; r) = \operatorname*{arg\,min}_{0<\Lambda<\mu(\tau;r)} \Big[K_\Lambda \Lambda^{-p} + K_\Delta(\mu(\tau; r) - \Lambda)^{-s}\Big],$$

and reduced objective

$$\Psi(\tau; r) = F(\tau, \Lambda^*(\tau; r); r).$$

9.2 The coupling penalty as a function of μ

Define

$$\phi(\mu) = \min_{0<\Lambda<\mu} \Big[K_\Lambda \Lambda^{-p} + K_\Delta(\mu - \Lambda)^{-s}\Big].$$

Then

$$\Psi(\tau; r) = (B_0 + B_1 r \tau^{-\gamma})\Big[K_0 + K_\tau \tau^q + \phi(\mu(\tau; r))\Big].$$

9.3 Lemma: small-μ asymptotics

Lemma 1 (Small-μ blow-up rate of ϕ)**.** *As $\mu \to 0$, one has*

$$\phi(\mu) \asymp C\,\mu^{-M}, \qquad M = \max(p, s),$$

for some constant $C > 0$.

Remark 1. When learner service rate μ is tiny, either the variance term or the staleness term has the harsher blow-up, and you can't scale it away by choosing Λ. The exact constant is not required for scaling arguments.

9.4 Exponent invariance regime

Assume an operating regime in which

$$\phi(\mu(\tau; r)) = o(\tau^q) \quad \text{and} \quad \frac{\partial}{\partial \tau}\phi(\mu(\tau; r)) = o(\tau^{q-1})$$

over the relevant range of (τ, r). This corresponds to "operate away from learner starvation so the optimized Λ penalty behaves like a bounded nuisance term."

Then the dominant balance in $\Psi(\tau; r)$ for the minimizer is between the increasing drift term $K_\tau \tau^q$ and the decreasing amortized comm term $r\tau^{-\gamma}$, yielding the same exponent:

$$\tau^*(r) \propto r^{1/(q+\gamma)}.$$

9.5 Coupling-dominated learner-starved regime

In a comm-dominated throughput regime,

$$t_{\text{step}}(\tau; r) \approx B_1 r \tau^{-\gamma} \quad \Rightarrow \quad \mu(\tau; r) \approx \frac{\tau^\gamma}{B_1 r}.$$

Using the lemma:

$$\phi(\mu(\tau; r)) \asymp C\left(\frac{\tau^\gamma}{r}\right)^{-M} = C\,r^M\,\tau^{-\gamma M}.$$

Thus the reduced objective contains a new competing power in τ introduced by coupling. Balancing the drift term $K_\tau \tau^q$ against this coupling-induced term gives the alternative scaling:

$$\boxed{\tau^*(r) \propto r^{\frac{M}{q+\gamma M}}, \qquad M = \max(p, s).}$$

Conclusion: coupling does not always preserve the exponent; invariance holds in the stable operating regime, while extreme comm-heavy regimes predict exponent shifts.

10 Gradient Normalization Semantics

This is the normalization relation used in the Valve discussion.

Assume each micro-step gradient $g_{j,k}$ is already averaged over a microbatch of size m on worker $j \in \{1, \ldots, W\}$ at local step $k \in \{1, \ldots, \tau\}$. Define the accumulated gradient on each worker

$$\text{accum}_j = \sum_{k=1}^{\tau} g_{j,k}.$$

Normalize by τW and take AllReduce mean to form the optimizer gradient

$$\bar{g} = \frac{1}{W} \sum_{j=1}^{W} \frac{1}{\tau} \sum_{k=1}^{\tau} g_{j,k} = \frac{1}{W\tau} \sum_{j,k} g_{j,k}.$$

This is exactly the "mean gradient per microbatch across all workers and τ steps" statement.

11 Summary of the General Scaling Law

The general scaling law consolidated is:

$$\tau^*(r) \propto r^{\alpha}, \qquad \alpha = \frac{1}{q+\gamma},$$

under the generalized exponent model

$$T_{\text{steps}}(\tau) \approx A_0 + A_1 \tau^q, \quad q > 1, \qquad t_{\text{step}}(\tau; r) \approx B_0 + B_1 r \tau^{-\gamma}, \quad \gamma > 0.$$

Special case $q = 2$, $\gamma = 1$ gives the cube-root law $\tau^*(r) \propto r^{1/3}$.